Tianyi Chen (Troy)

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EDUCATION

University of Michigan, Honors College of Engineering — Ann Arbor, MI
B.S.E – Mechanical Engineering, Minor – Artificial Intelligence, GPA: 3.7 — May 2028
Organizations/Awards: Honors Program, Dean's List, ASME, UMEB

EXPERIENCE

University of Michigan – Walter E. Lay Automotive Lab — Ann Arbor, MI
RESEARCH ASSISTANT – Combustion Engine Efficiency R&D — September 2026

- Designing PCB and building a photodiode-based sensor using transimpedance amplifiers capable of transmitting signals from 8 photodiode to LabVIEW, for visualizing flame propagation in combustion chamber
- Fueling diesel engines with 100% ethanol, modifying 4-stroke cycle to reduce CoV while increasing efficiency
- Refining exhaust rebreathing to increase piston temperature for stable combustion of ethanol when idling

University of Michigan – Zoetic Robotics Lab — Ann Arbor, MI
RESEARCH ASSISTANT – Controlled Micro-Combustion R&D — September 2026

- Testing insect scale micro-actuator driven via controlled combustion, targeting high force output and controllable locomotion through fuel oxidizer ratio, ignition frequency control, and actuation timing
- Developing a portable fuel tank capable of performing a controlled redox process to produce fuel for the combustion actuator while managing by-products, eliminating dependency on nearby stationary fuel sources
- Exploring and testing new solutions of signal transport via method of precisely controlled combustion reaction

Stellbot (星际智行) – Design Research Team — Nanjing, China
MECHANICAL ENGINEERING INTERN – Hardware/Structure Design and Testing — June 2026 - August 2026

- Rearranged and Converted complex robotic dog SolidWorks file into a Unified Robot Description Format file (URDF) in preparation to use in NVIDIA Isaac Sim for reinforcement learning
- Used Easy EDA, a PCB design software to design circuit boards for robotic dogs, contributions include a port hub for all 12 joint motors and an STM32 powered PCB for robot internal hardware environment monitoring
- Debugged robot dog lithium battery pack code by making a monitor app using Python that displayed battery data along with processing battery usage data automatically to catch unusual behavior and errors when undergoing charging with different current and uneven discharging

Michigan State University – Spence Lab — East Lansing, MI
RESEARCH ASSISTANT – Medical Research with RBC — August 2024 - May 2025

- Studied effects of C-peptide in patients with type one diabetes by measuring ATP release of red blood cells using a dual pump injection system, luciferin luciferase mixture, and photodiode sensor
- Explored studies involved use of normoglycemic blood storage solution for red blood cell storage, aims to decrease rate of red blood cell deformation and cell lysis, increasing success of blood transfusion

PROJECTS AND TEAMS

Smart Watch: Designed a simple smart watch — June - August 2026

- Designed and manufactured PCB for a smartwatch using Easy EDA with hardware for two-pin magnetic charging, temperature, humidity, and heart rate monitoring using a STM32 MCU
- Programmed smart watch PCB and touch screen display using Hardware Abstraction Layer (HAL) library with CubeMX and Keli µvision capable of displaying various backgrounds, time, temperature and humidity

Design and Manufacturing I: Designed robotic RMP, University of Michigan COE — January - May 2026

- Worked in a student team designing a robot on SolidWorks capable of picking objects from ground in less than 10 seconds utilizing a virtual-four-bar claw
- Manufactured and assembled all parts of the robot for testing and improvements, before entering competition

TECHNICAL SKILLS

Manufacturing: Mill, Lathe, Bandsaw, Waterjet, 3D/Resin Printing, Bill of Materials, Assembly, Soldering PCB
Computer: Solid Works, NX, Fusion 360, LabVIEW, Python, C++, Easy EDA, Adams, Photoshop and Illustrator
Hobbies: Fencing, Badminton, volleyball, camping, hiking, trying out a bunch of different foods from around the world